

Collective Robotics

Part 7: Case study – adaptive aggregation

Prof. Dr. Javad Ghofrani



**Hochschule
Bonn-Rhein-Sieg**
University of Applied Sciences

Recap

- One sentence about what you have learned last time?
- Explanation of the decision making under uncertainty

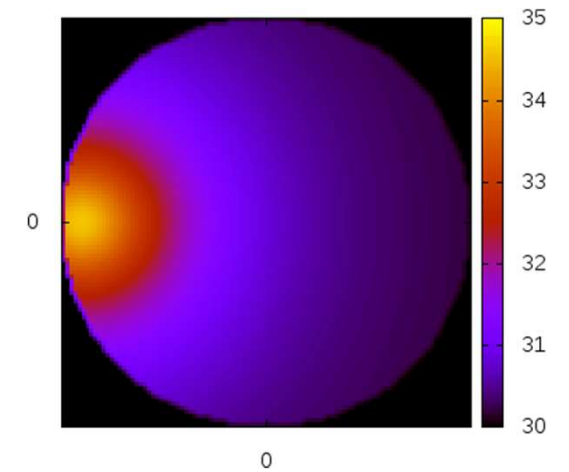
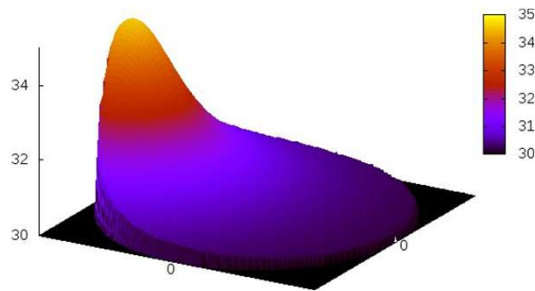
A swarm robotics use case...

Say, a customer orders **software** that controls a **robot swarm** which aggregates at a certain spot determined by sensor input and stays **flexible** to a dynamic environment.



AI-Generated

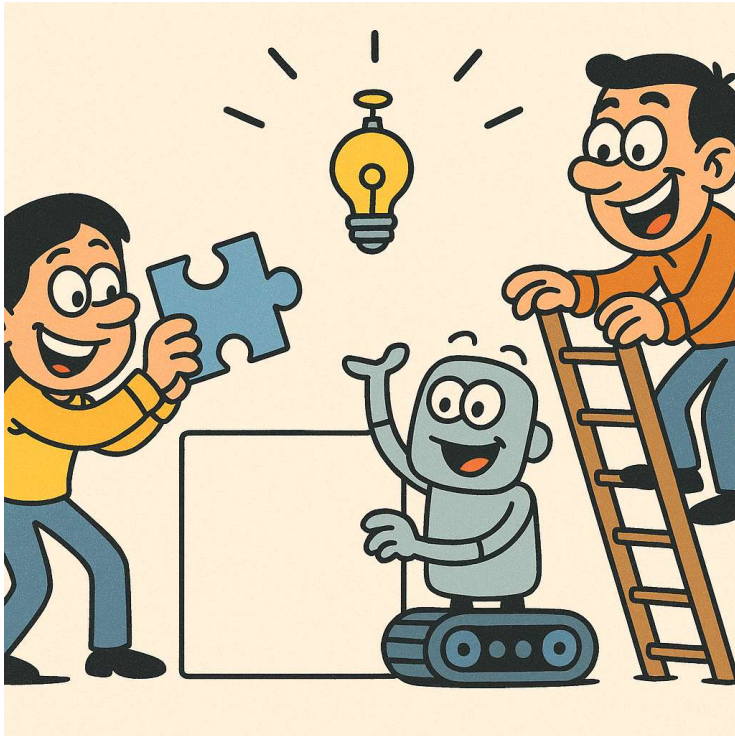
Aggregation at warmest spot



More complex possibilities:

- local optima
- scattered spots systematic
- plateaus etc.

Alternative solutions 1: ad-hoc approach



Simple ad-hoc approach (no-brainer?)

idea:

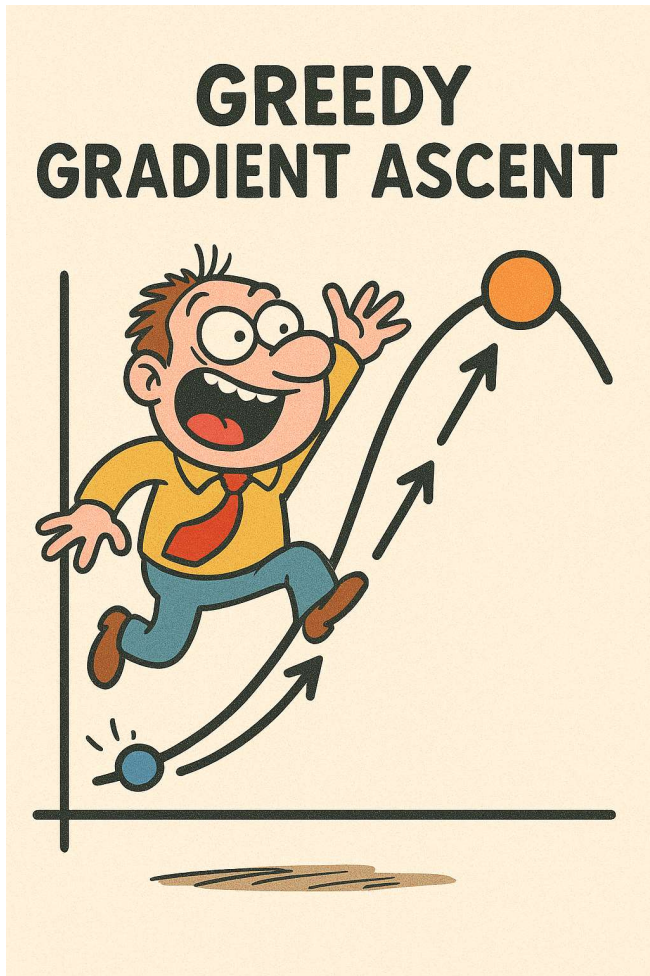
- drive randomly around while measuring
- keep track of biggest measured values
- after some time, stop when approaching big values again

Ignores multi-robot setting

How to make use of swarm members?

- for example, share biggest measured values

Alternative solutions 2: gradient ascent



Greedy gradient ascent requirements:

- smooth gradient, no local maxima
- reliable sensors
- measurement of gradient either with 2 sensors
- or over time

Ignores multi-robot setting

How to make use of swarm members?

- communicating promising directions could be an option

Alternative solutions 3: positive feedback

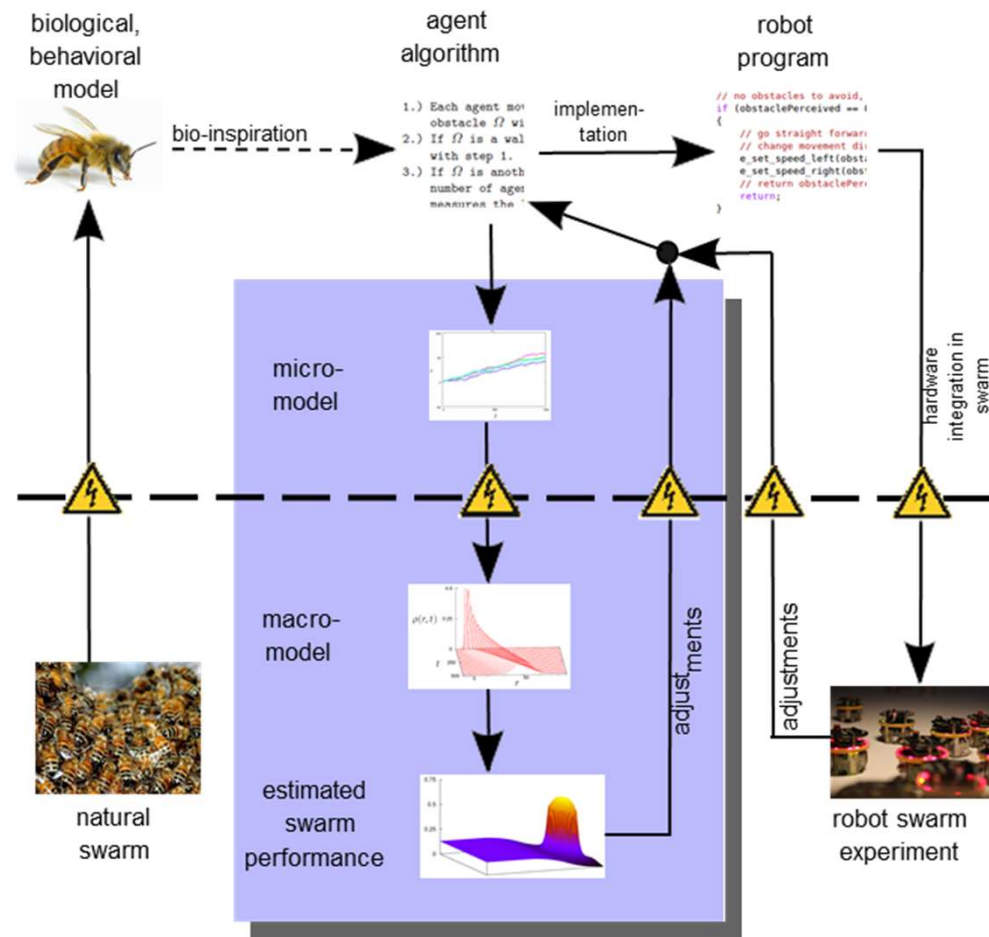


Positive feedback approach (follow the others)

idea:

- search for a good spot
- hang around
- if no one joins, give up,
- and try finding another spot

Micro-macro models



Biological inspiration

aggregating young honeybees



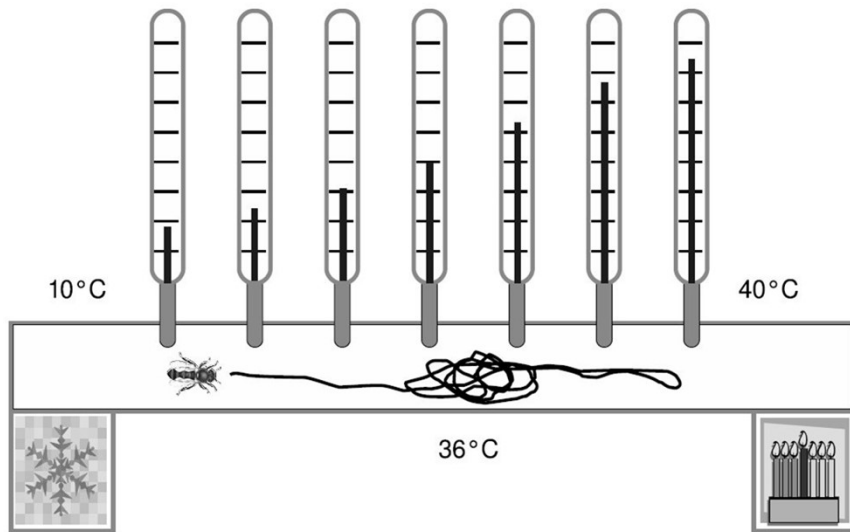
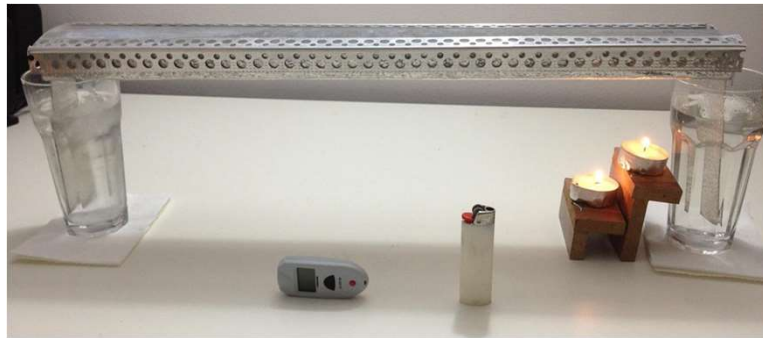
Young (<24 hours)

Cannot fly nor sting, only walk

Prefer certain temperature (36° C, to dry wings) known to show reactions to social stimuli

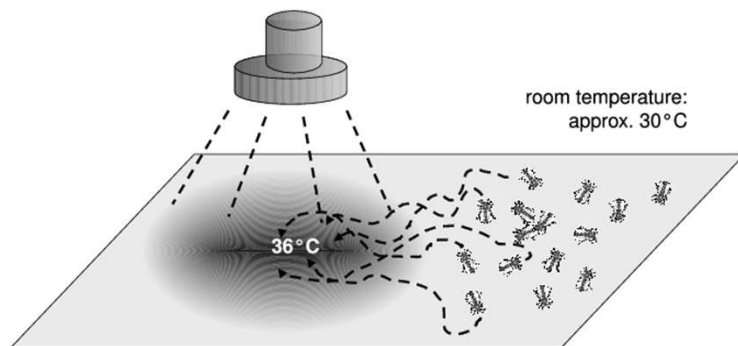
Stay in the hive (dark, regions with different temperatures) observation: aggregate at spots of 36° C

Honeybee behavior in “pseudo-1-d”



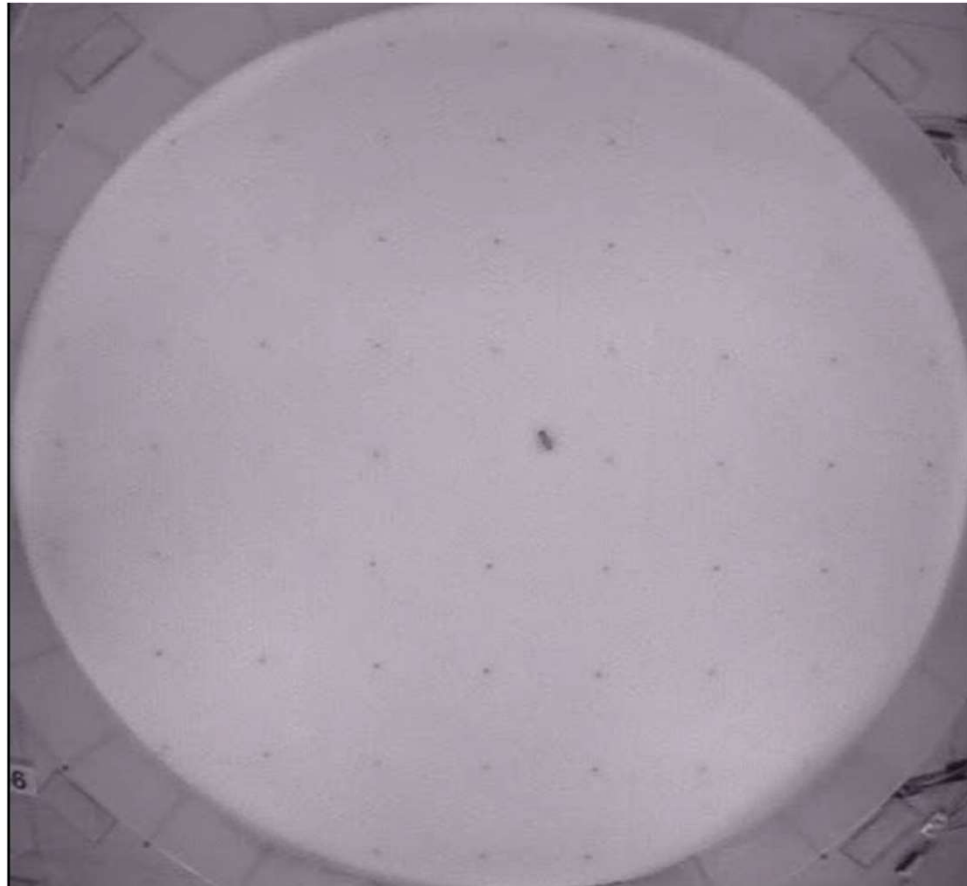
single bee finds temperature optimum

Honeybee behavior in 2-d + “flat gradient”

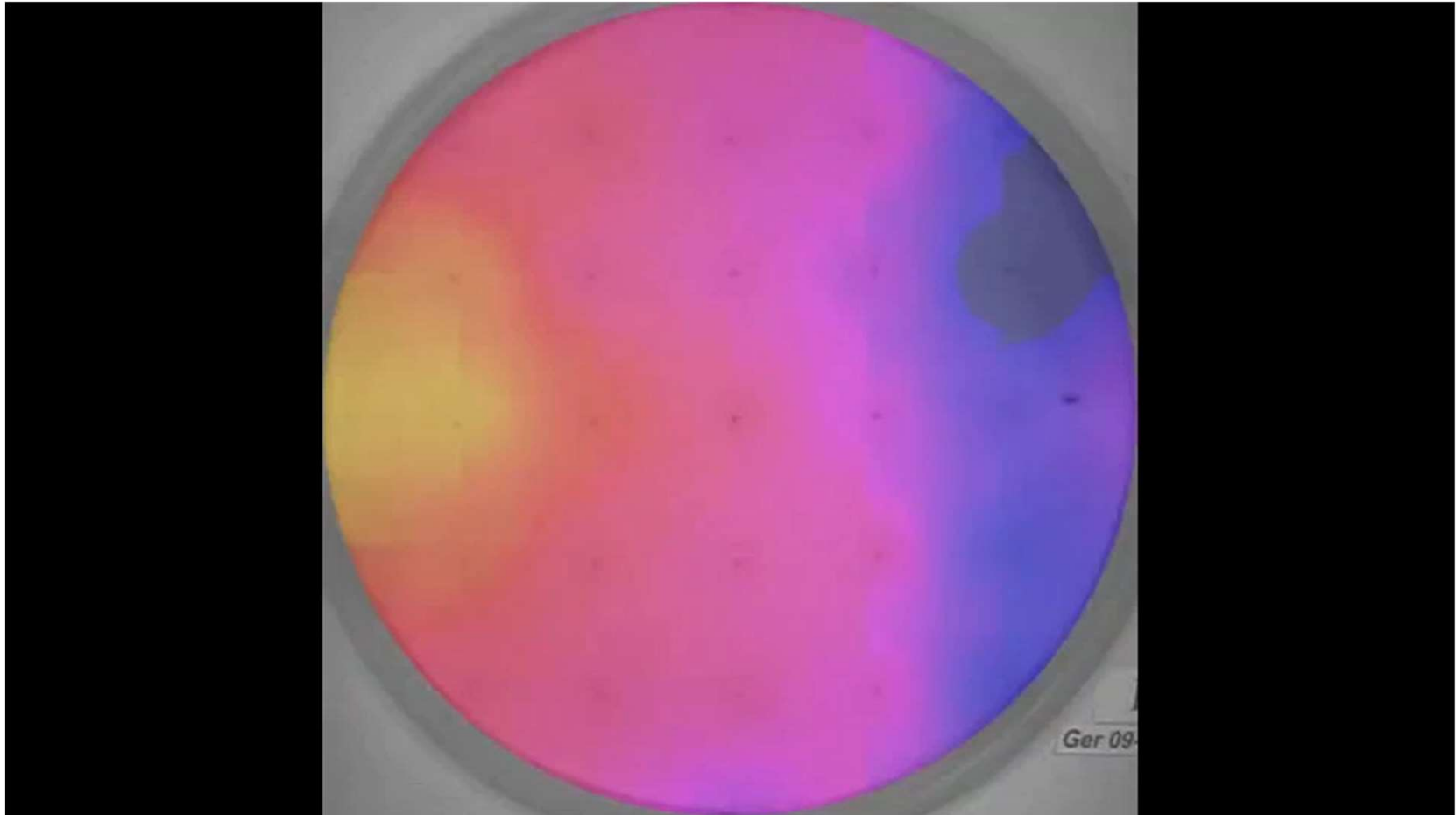


- round arena (in contrast to the above image) covered with honeycombs
- all dark
- optimal (36° C) spot on left side recording with infrared camera
- array of temperature sensors in floor below honeycombs bee(s) are initially placed in the center

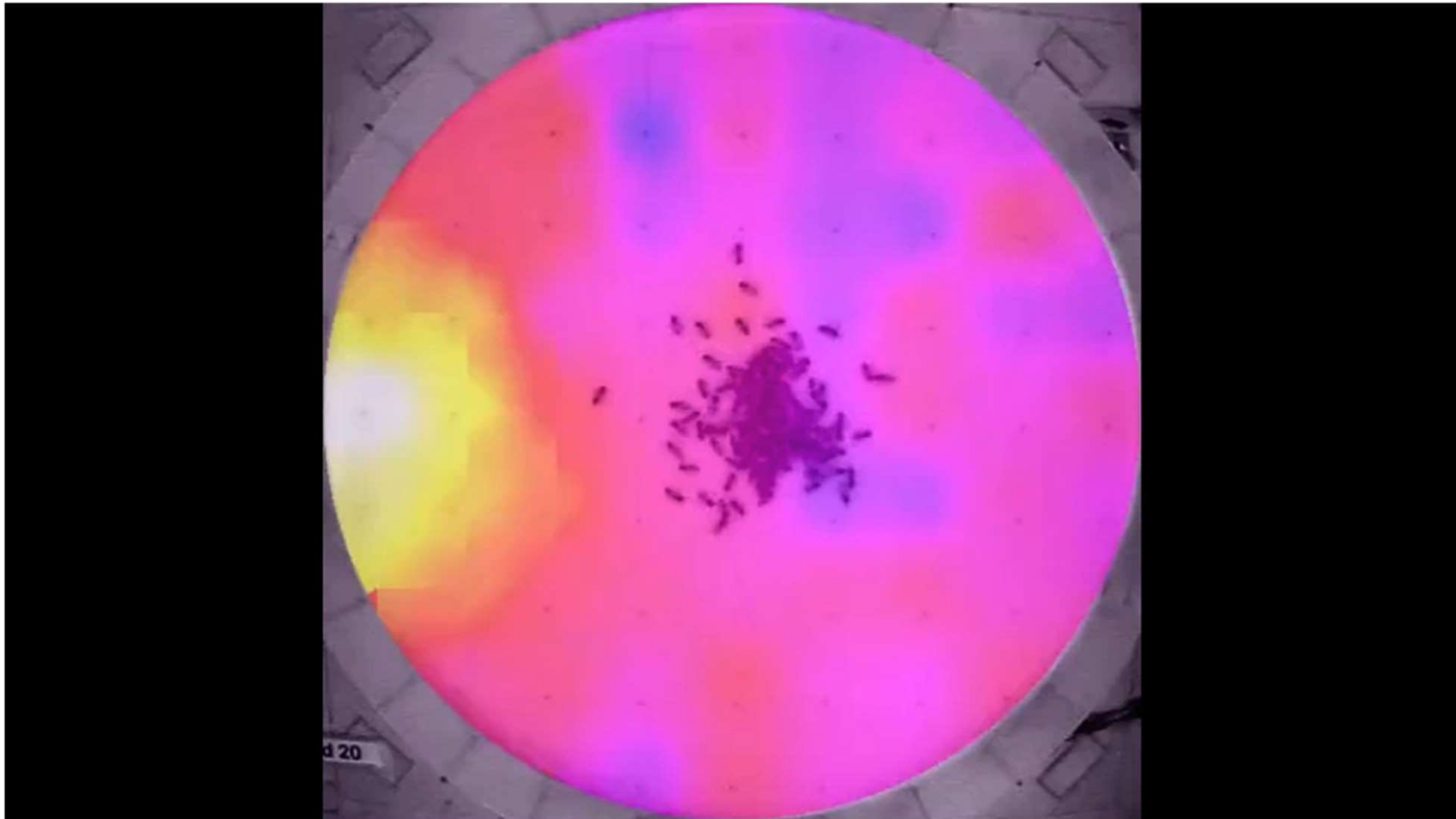
Honeybee behavior – single bee



Honeybee behavior – single bee



Honeybee behavior – group of bees



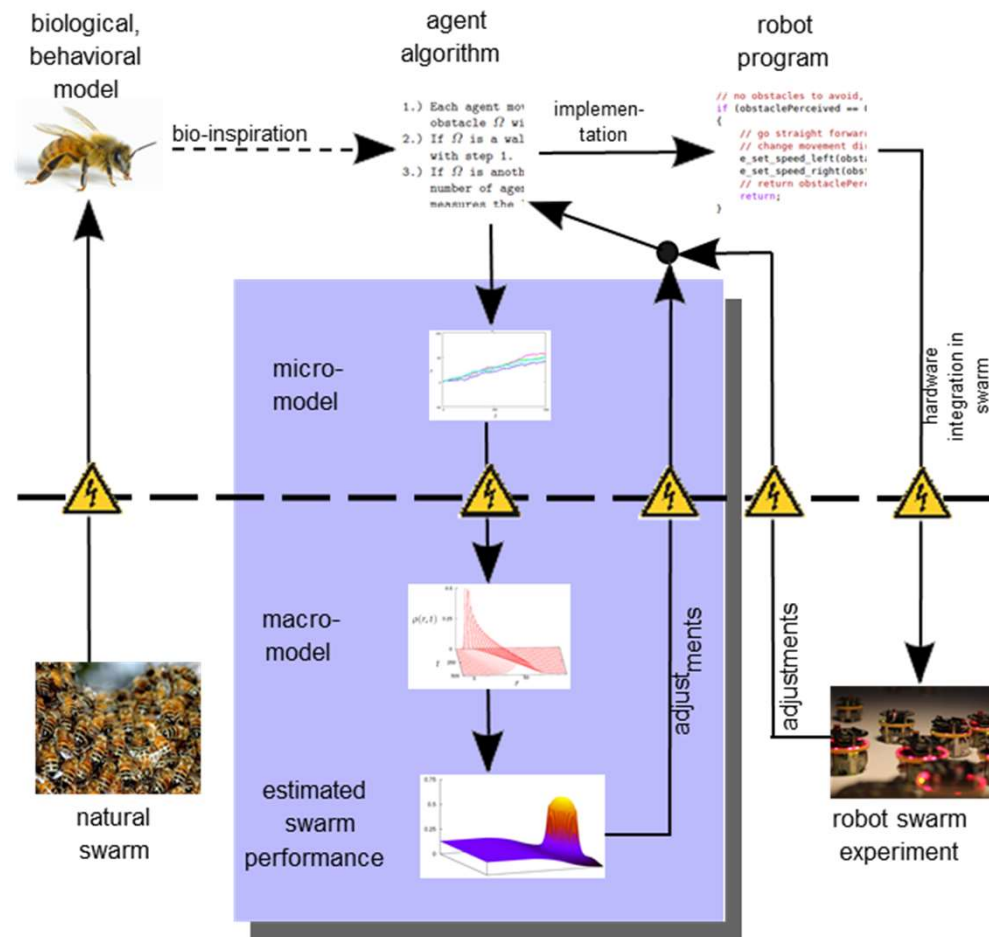
Aggregating honeybees

- single bee cannot find optimum
 - few bees are inefficient in finding the optimum many bees find the optimum efficiently
 - swarm effect: better individual(!) performance with increasing group size
- ⇒ hypothesis:



some social interaction is a crucial ingredient of this behavior

Micro-macro models



Behavioral model

No known reductionist approach for determining behavior (i.e., “cutting the bee brain into pieces does not help”)

Behavior model is a hypothesis

1. move straightforward
2. obstacles or robots around?
3. in case of an obstacle: turn away, return to step 1
4. in case of robot:
 - stop,
 - measure temperature,
 - wait for some time (the longer the higher the temperature),
 - U-turn,
 - return to step 1

(variation: wait for shorter time for very high temperatures)

Micro-macro modeling approach

In the following we want to create
a microscopic model for an individual robot and a macroscopic model for
the whole swarm based on the Langevin equation
and the Fokker–Planck equation.

Modeling objectives:

- represent difference between ineffective single behavior in flat gradient and effective behavior of swarm
- try to represent parameters of the agent algorithm directly in the model
- represent spatial features
- quantitative agreement with experiments

1st approach to micro model

We use a Langevin equation to model the individual behavior with gradient ascent depending on parameter α :

$$\dot{\mathbf{R}}(t) = \alpha \nabla P(\mathbf{R}(t)) + \mathbf{B}\mathbf{F}(t)$$

$\mathbf{R}$ robot position (here in 2-d)

P temperature field

$\nabla P(\mathbf{R}(t))$ gradient of temperature field at robot's current position

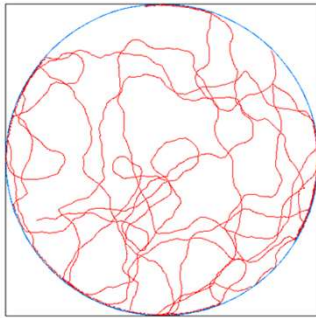
$\alpha \in [0,1]$ intensity of goal-oriented behavior

$\mathbf{F}$ stochastic process that models random behavior (walking in random directions)

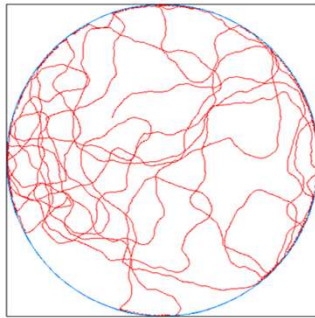
B intensity of random behavior

Sampled trajectories

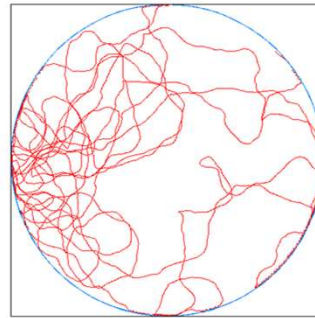
$\alpha = 0.05$



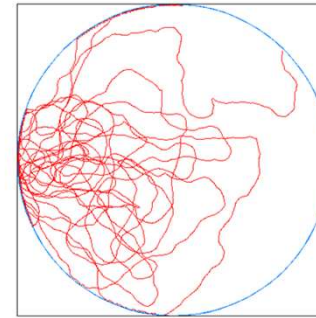
$\alpha = 0.1$



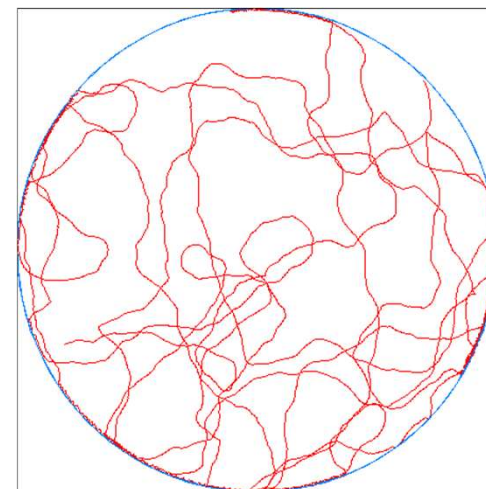
$\alpha = 0.15$



$\alpha = 0.2$



Micro model (1/3)



For the flat-gradient setting as occurring in the experiments we set $\alpha = 0$ and turn the goal-oriented behavior of a single bee off

$$\dot{\mathbf{R}}(t) = \alpha \nabla P(\mathbf{R}(t)) + \mathbf{BF}(t) = \mathbf{BF}(t)$$

We identify two states: a robot is either *moving* or *stopped*.

We define a function which gives the corresponding waiting time for a given position $\mathbf{R}$

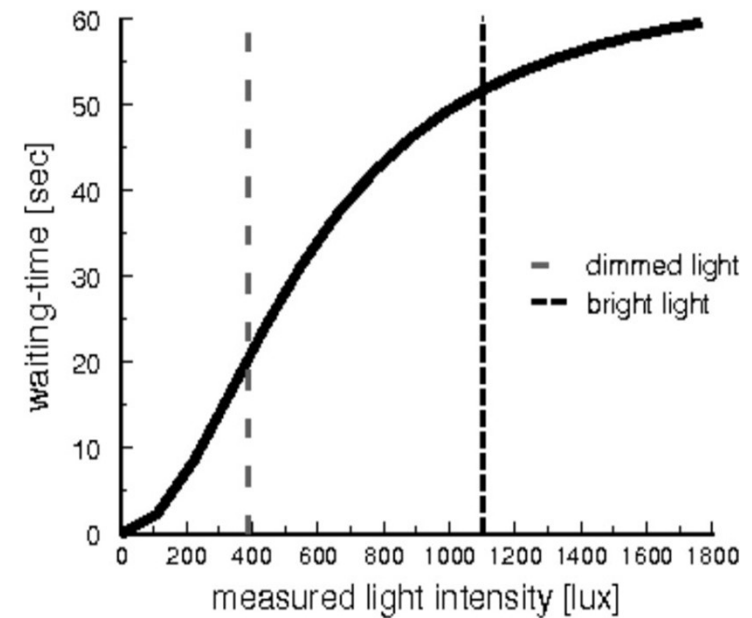
(and the associated temperature $P(\mathbf{R})$).

Micro model (2/3)

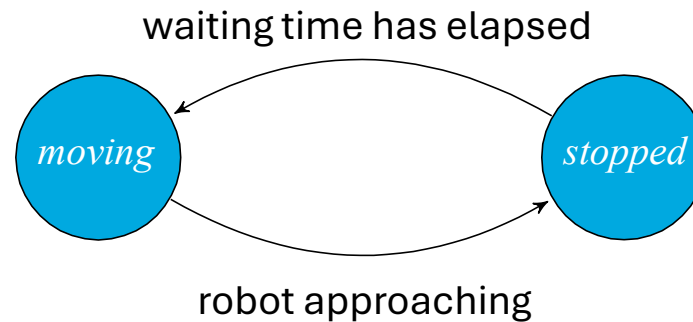
Waiting time function $w(\mathbf{R}) = \frac{w_{\max} P^2(\mathbf{R})}{P^2(\mathbf{R}) + c}$ (2)

for maximal waiting time $W_{\max}$, temperature $P(R)$ at position R , and scaling constant c .

An example of a waiting time function for light intensity instead of temperature:



Micro model (3/3)



motion in state *moving*:

$$\dot{\mathbf{R}}_m(t) = \mathbf{BF}(t) \quad (3)$$

transitions to stopped based on encounters of other robots, motion in state *stopped*:

$$\dot{\mathbf{R}}_s(t) = 0 \quad (4)$$

transitions back to moving based on a waiting time $w(R)$

Macro model (1/4)

the macroscopic counterparts to the micro model

$\rho_m(\mathbf{r}, t)$ density of moving robots within an infinitesimally small area located at $\mathbf{r}$ at time t

$\rho_s(\mathbf{r}, t)$ density of stopped robots at $\mathbf{r}$ at time t

model of motion in state *moving*:

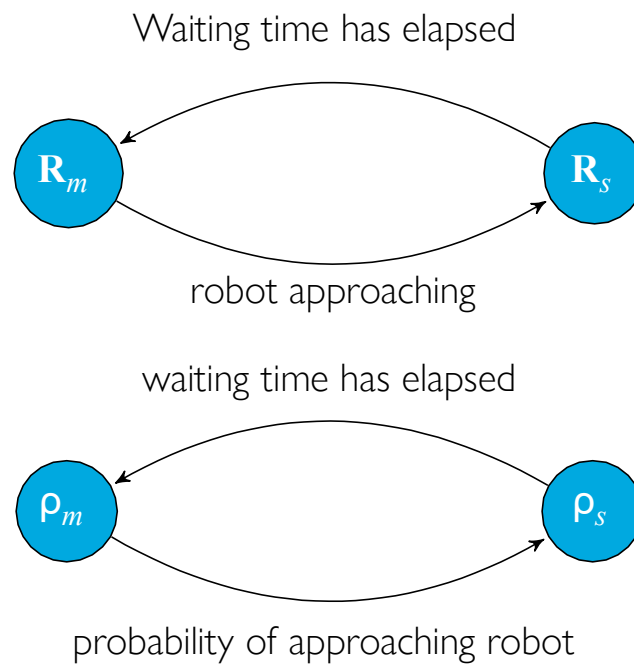
$$\dot{\mathbf{R}}_m(t) = \mathbf{B}\mathbf{F}(t) \Rightarrow \frac{\partial \rho_m(\mathbf{r}, t)}{\partial t} = B^2 \nabla^2 \rho_m(\mathbf{r}, t) \quad (5)$$

model of motion in state *stopped*:

$$\dot{\mathbf{R}}_s(t) = 0 \Rightarrow \frac{\partial \rho_s(\mathbf{r}, t)}{\partial t} = 0 \quad (6)$$

Macro model (2/4)

the macroscopic counterparts to the micro model

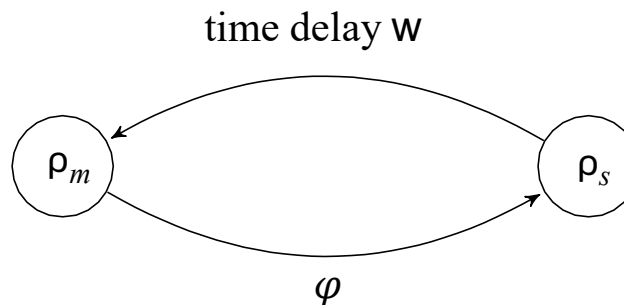


Macro model (3/4)

determining the transitions

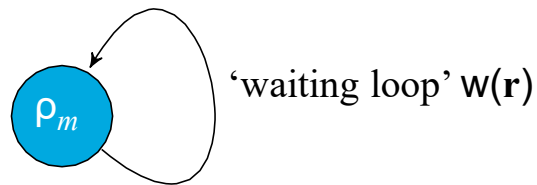
The discrete event of one robot approaching another robot does not exist anymore for robot densities. Instead we have only a **probability of meeting a robot** and stopping.

This is modeled by a stopping rate φ .



Macro model (4/4)

ρ_s is mathematically unnecessary
we delete it by using time delays instead



The swarm fraction that is subtracted from $\rho_m(\mathbf{r}, t_0)$ at time t_0 is added to $\rho_m(\mathbf{r}, t_1)$ at time $t_1 = t_0 + w(\mathbf{r})$

Complete macro model

density of robots in state moving

$$\frac{\partial \rho_m(\mathbf{r}, t)}{\partial t} = B^2 \nabla^2 \rho_m(\mathbf{r}, t) - \rho_m(\mathbf{r}, t) \varphi + \rho_m(\mathbf{r}, t - w(\mathbf{r})) \varphi$$

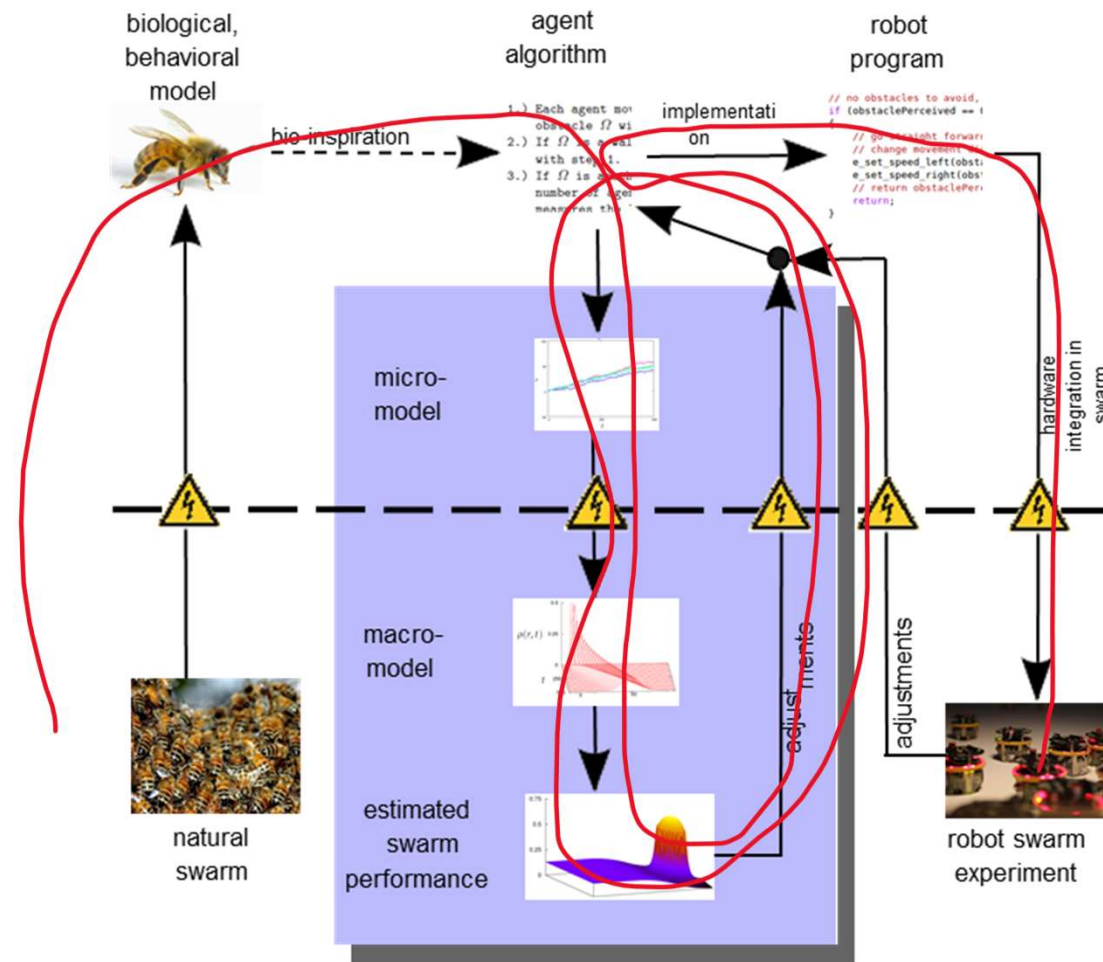
(φ is stopping rate)

not necessary but for a better understanding:

density of robots in state *stopped* (*has no motion*)

$$\frac{\partial \rho_s(\mathbf{r}, t)}{\partial t} = \rho_m(\mathbf{r}, t) \varphi - \rho_s(\mathbf{r}, t - w(\mathbf{r})) \varphi$$

Micro-macro models



Validation in experiment

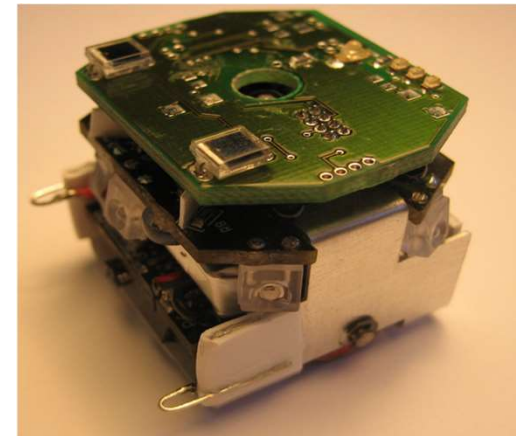
15 Jasmine robots

arena 150 cm × 100 cm

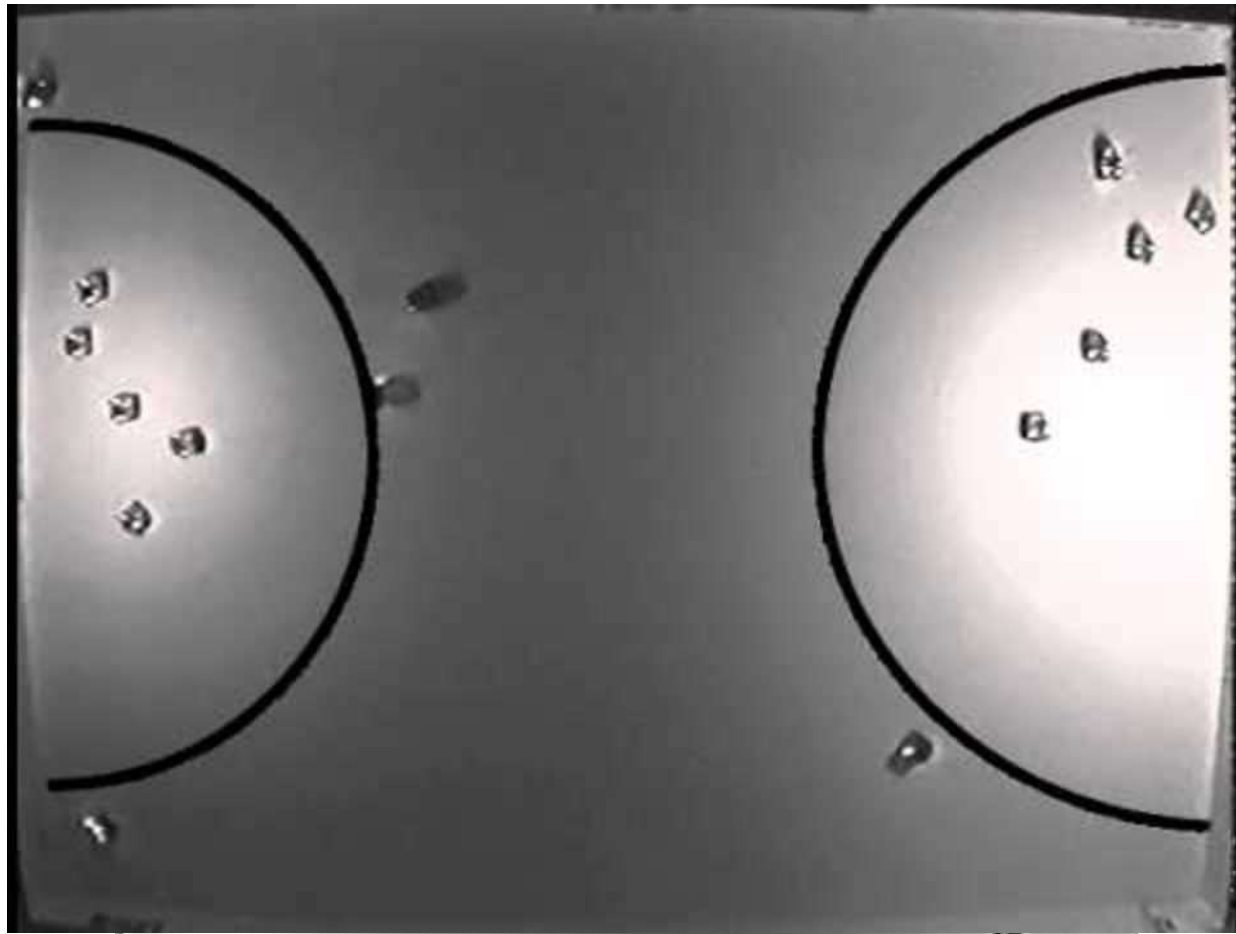
one lamp at each of the two short sides of the arena

lamps are operated in three modes: off, dimmed and bright the experiment has 4 phases (left/right):

- off, dimmed
- bright, dimmed
- dimmed, bright
- dimmed, off



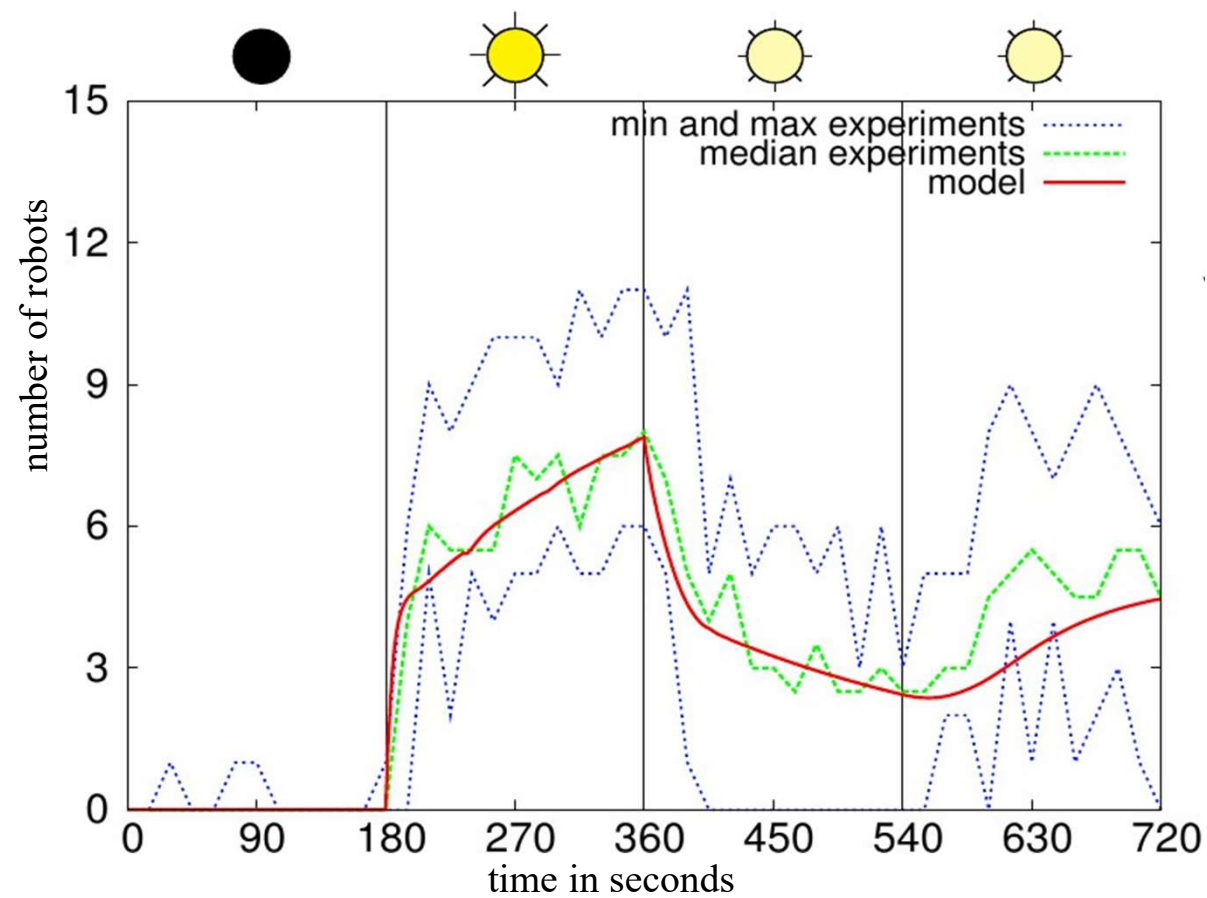
Videos: model and robot swarm



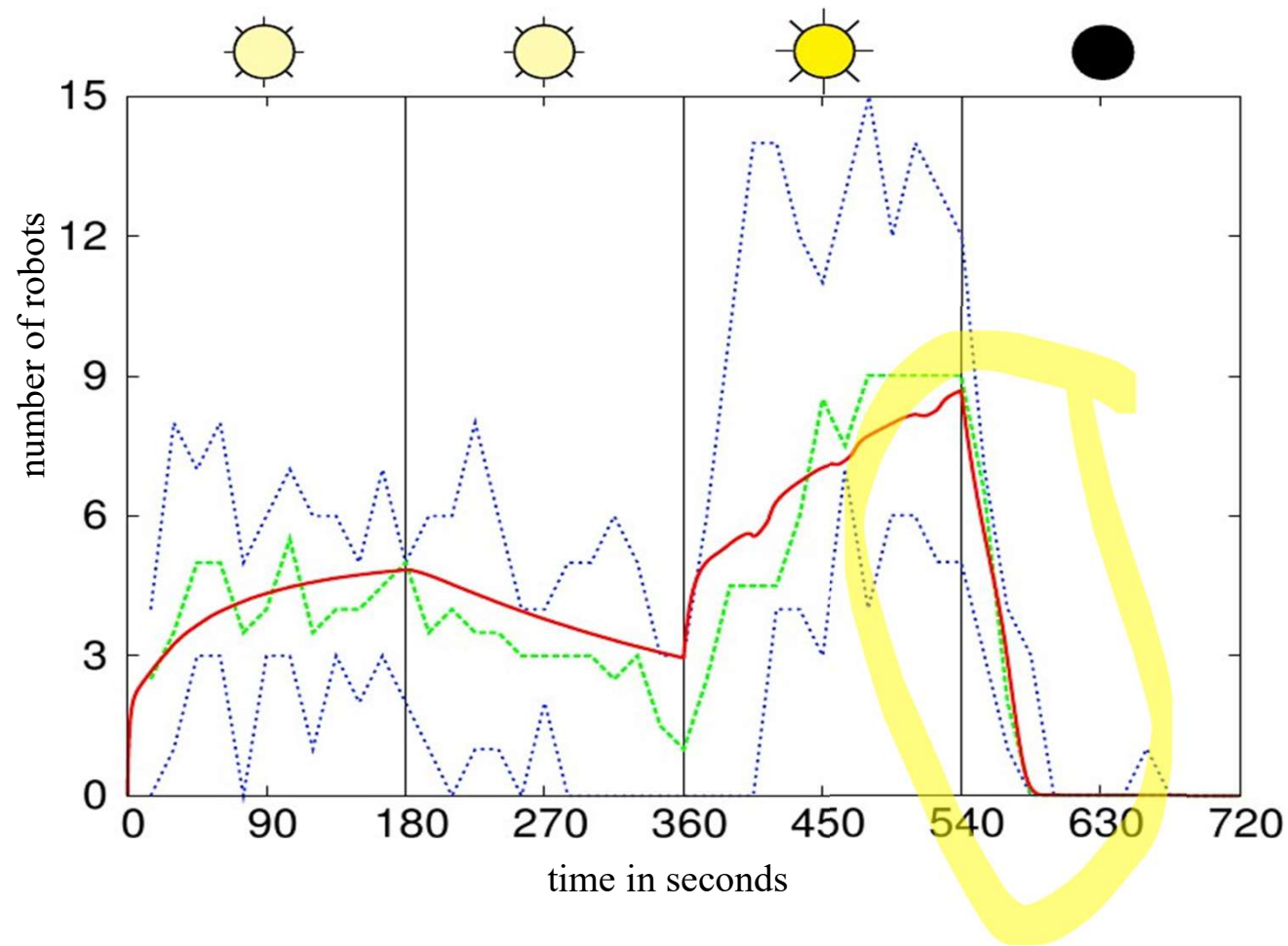
Question!

- What is the biggest challenge in this system?

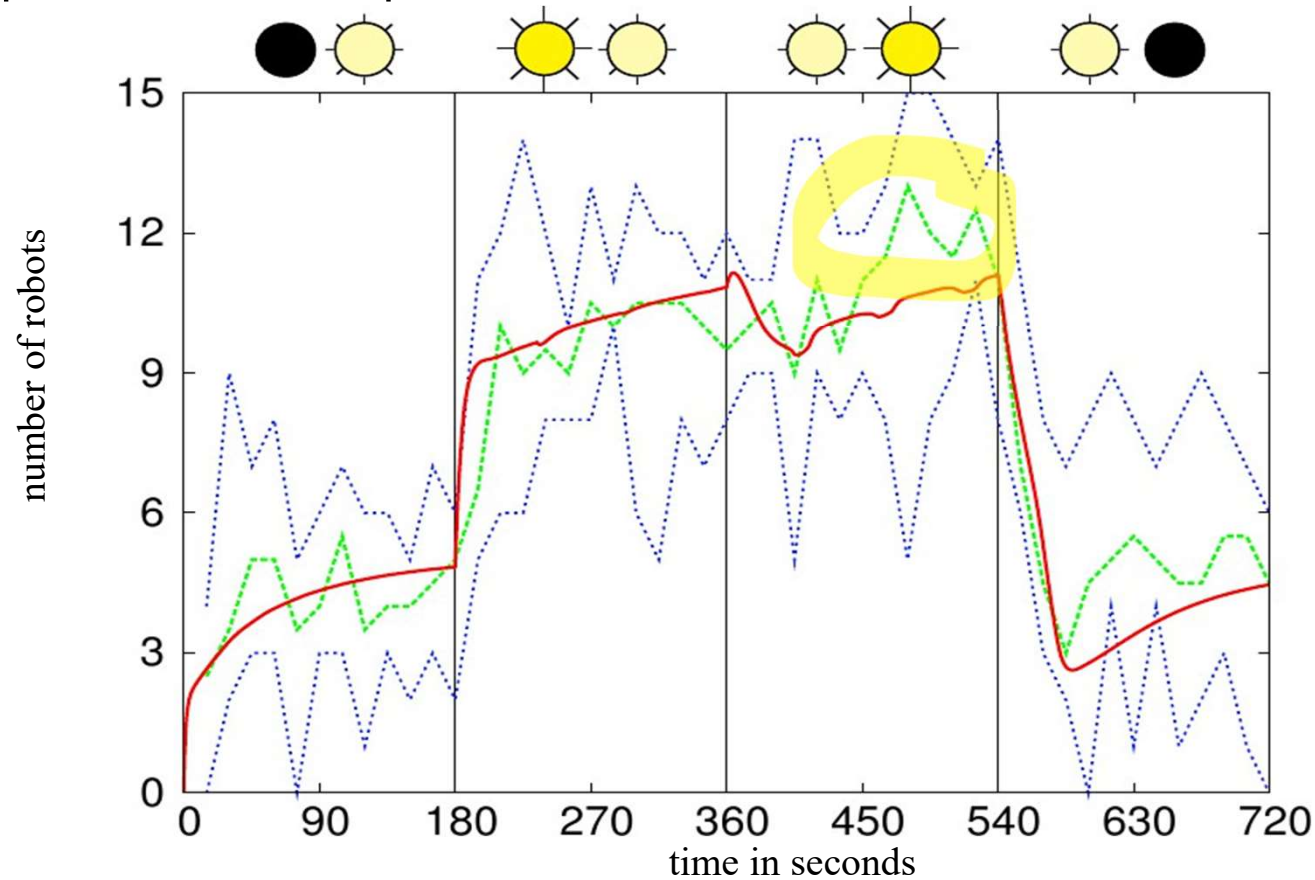
Camparison Experiment-Model – aggregation Left



Comparison Experiment-Model – Aggreg. Right



Comparison Experiment-Model – Total Aggreg.

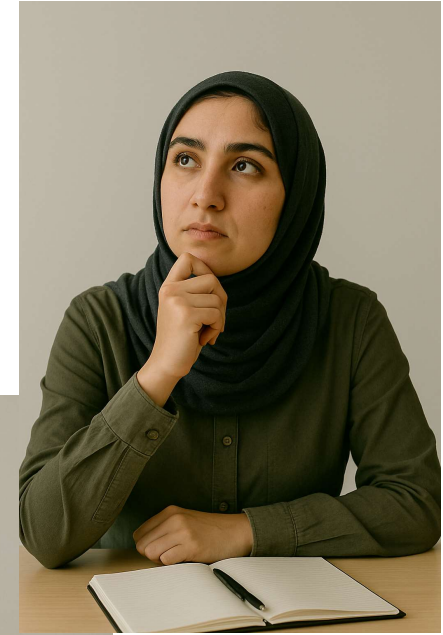


Conclusion

- aggregation in 'flat' temperature fields in honeybees is a swarm effect
bio-inspiration helped in defining an effective algorithm
- the swarm is adaptive to a dynamic environment
- the model is in good agreement with the robot experiments
- the model can be used to optimize the waiting time function and the stopping rate
- let's hope our customer is satisfied :-)

Two minutes of thinking

- Think one minute and write one sentence
one minute for highlight
one minute for challenge



Reference

- Schmickl, Thomas, and Heiko Hamann. "BEECLUST: A swarm algorithm derived from honeybees." *Bio-inspired computing and communication networks* (2011): 95-137.
- Pdf available here:
https://heikohamann.de/pub/schmickl_beeclust_2011.pdf